Causal Inference with Large Language Models: Identification, Measurement, and Estimation

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Abstract

Establishing causation rather than mere correlation remains the central challenge of empirical social science. This manuscript provides a rigorous, mathematically formalized framework for integrating large language models (LLMs) into the causal inference pipeline. We explore how LLMs serve as measurement instruments to extract latent constructs from unstructured text, augmenting the Neyman-Rubin potential outcomes framework and Pearl's structural causal models. We define the conditions under which design-based supervised learning (DSL) yields consistent estimators of the average treatment effect despite non-random LLM measurement error. We further formalize the role of LLMs in causal discovery, treating them as prior probability distributions over directed acyclic graphs (DAGs) to resolve Markov equivalence classes. Moving to estimation, we derive doubly robust estimators under inference-time text confounding, proving Neyman orthogonality conditions for double/debiased machine learning. Finally, we apply counterfactual auditing to LLM fairness, defining causal bias metrics and detailing their estimation. This manuscript synthesizes recent methodological advances, offering a unified, mathematically precise perspective on the intersection of generative AI and causal inference.

1 Introduction

Establishing causation, rather than mere correlation, remains the central challenge of empirical social science. The fundamental difficulty arises from the impossibility of observing counterfactual outcomes simultaneously. Let $i \in \{1, \ldots, N\}$ index independent and identically distributed units. Let $D_i \in \{0, 1\}$ denote a binary treatment, and let $Y_i(1)$ and $Y_i(0)$ denote the potential outcomes under treatment and control, respectively [Rubin, 1974]. The observed outcome satisfies the switching equation:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0). \tag{1}$$

The individual causal effect $\tau_i = Y_i(1) - Y_i(0)$ is not directly observable because one of the two potential outcomes is always missing. This fundamental missing data structure implies that causal

inference cannot proceed without additional assumptions linking the observed data to the counterfactual quantities of interest.

Consequently, empirical analysis focuses on population-level estimands such as the Average Treatment Effect (ATE):

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)]. \tag{2}$$

The identification and estimation of τ require a combination of design assumptions, structural restrictions, and statistical procedures. Over the past decades, two complementary frameworks have emerged as the dominant paradigms for addressing these challenges. The potential outcomes framework formalizes causal inference as a problem of missing data, while Structural Causal Models (SCMs) provide graphical and structural tools for reasoning about identification [Pearl, 2009].

In parallel, the increasing availability of large-scale unstructured data has led to the development of text-as-data methodologies [Gentzkow et al., 2019]. These approaches typically rely on dimensionality reduction techniques or predictive models to extract structured information from text. However, the majority of such methods have been designed for prediction rather than causal inference. As a result, they often lack explicit identification strategies and may inadvertently introduce biases when used in causal settings.

Large Language Models (LLMs) introduce a qualitative shift in this landscape. Because LLMs are pre-trained on vast corpora, they possess a zero-shot capability to map high-dimensional text sequences $\boldsymbol{W}_i \in \mathcal{V}^*$ (where $\mathcal{V}$ is a vocabulary) to lower-dimensional, latent semantic spaces $\boldsymbol{Z}_i \in \mathbb{R}^k$ encoding both semantic and contextual information. Unlike traditional feature extraction techniques, LLMs can generate flexible, task-specific representations without requiring explicit supervision for each downstream task. This capability suggests that LLMs can serve as high-throughput measurement devices, transforming unstructured text into variables that can be incorporated into causal inference pipelines.

However, this integration raises fundamental methodological questions. LLM-generated outputs are not classical measurements: they are subject to systematic, context-dependent, and potentially non-random errors. These errors may depend on both observed and unobserved characteristics of the data-generating process, thereby violating standard assumptions used in causal inference. Consequently, naive substitution of LLM outputs into causal estimators may lead to biased or inconsistent results.

This manuscript formalizes the integration of LLMs into the causal inference pipeline. We do not merely propose LLMs as heuristic tools; rather, we embed their outputs into rigorous statistical frameworks. We define the conditions under which LLM-generated surrogates can be used for valid downstream inference, derive the asymptotic properties of doubly robust estimators utilizing LLM embeddings, and formalize the use of LLMs as structural priors in causal discovery algorithms.

The contributions of this paper are as follows. After reviewing the foundational literature, first

we formalize LLMs as measurement instruments and characterize the conditions under which their outputs can be used in downstream causal analysis. Second, we establish the consistency properties of design-based supervised learning estimators in the presence of non-random measurement error. Third, we interpret LLMs as sources of prior information in causal discovery and formalize their role within Bayesian structure learning. Fourth, we derive doubly robust estimators in settings where both treatments and confounders are extracted from text. Finally, we define counterfactual fairness metrics for auditing LLM outputs using causal principles. By integrating generative AI into the formal machinery of causal inference, this manuscript aims to provide a unified theoretical foundation that supports both methodological rigor and practical applicability.

2 Literature Review

The intersection of causal inference and natural language processing has developed rapidly in recent years, driven by the increasing availability of text data and advances in machine learning. Early contributions focused on representing text in a form suitable for statistical analysis. Gentzkow et al. [Gentzkow et al., 2019] provided a comprehensive framework for treating text as data, emphasizing dimensionality reduction techniques such as topic models and regularized regression. While these approaches enabled the incorporation of text into econometric models, they were primarily designed for predictive tasks and did not directly address issues of causal identification.

A key development in this direction was the work of Egami et al. [Egami et al., 2022], who introduced a formal framework for causal inference with text-derived variables. They distinguished between several roles that text can play in causal analysis, including text as a treatment, as an outcome, and as a confounder. Their analysis demonstrated that naive application of supervised learning to extract treatments or outcomes from text can induce post-treatment bias, particularly when the training procedure conditions on variables that are themselves affected by the treatment. To address this issue, they proposed design-based strategies that separate the measurement stage from the estimation stage.

More recent work has expanded these ideas in the context of large language models. Surveys by Ma et al. [Ma et al., 2024] and Liu et al. [Liu et al., 2025] categorize the emerging literature into two broad directions: the use of LLMs for causal inference tasks, and the use of causal inference methods to analyze and audit LLM behavior. The former includes applications such as treatment effect estimation, causal discovery, and variable construction, while the latter focuses on issues of fairness, bias, and interpretability.

In causal discovery, traditional algorithms such as the PC algorithm [Spirtes and Glymour, 1991] rely on conditional independence tests to recover the structure of a directed acyclic graph. Under the assumptions of the Causal Markov condition and faithfulness, these methods are asymptotically consistent but can only identify the true graph up to its Markov Equivalence Class (MEC). Recent

work has explored the use of LLMs as sources of prior information to orient edges within equivalence classes. Khatibi et al. [2024] introduced the Autonomous LLM-Augmented Causal Discovery (ALCM) framework, formalizing the integration of LLM priors to orient edges within the MEC, while Darvariu et al. [2024] demonstrated the empirical efficacy of this approach. While this approach can improve empirical performance, it introduces new challenges related to the reliability and interpretability of the learned priors.

For causal estimation in high-dimensional settings, Double/Debiased Machine Learning [Chernozhukov et al., 2018] provides a generalized framework for obtaining root-N consistent estimates of low-dimensional causal parameters in the presence of high-dimensional nuisance parameters estimated via machine learning. This builds upon the foundational semiparametric efficiency theory of Robins et al. [1994]. By exploiting Neyman orthogonality, DML estimators remain robust to first-order errors in nuisance function estimation. This property is particularly relevant in settings involving text data, where nuisance functions must be estimated using flexible, high-capacity models. The theoretical foundations of this approach build on earlier work in semiparametric efficiency [Robins et al., 1994]. Recent work [Ma et al., 2025] has extended DML to accommodate inference-time text confounding, proving that LLM embeddings can serve as valid adjustment sets under specific conditional independence assumptions.

Despite these advances, a unified theoretical framework that systematically incorporates LLMs into all stages of the causal inference pipeline remains lacking. In particular, existing work often treats LLM outputs as given, without explicitly modeling their measurement properties or their interaction with identification assumptions. This manuscript addresses this gap by providing a formal treatment of LLMs as components of the causal inference process, with explicit attention to identification, measurement error, and estimation.

3 Formal Foundations of Causal Inference

To integrate LLMs rigorously, we must formalize the identification conditions under the potential outcomes framework.

Assumption 1 (Stable Unit Treatment Value Assumption (SUTVA))**.** *Let $\boldsymbol{D} = (D_1, \ldots, D_N)$ be the vector of treatment assignments for all units. SUTVA requires that $Y_i(\boldsymbol{D}) = Y_i(D_i)$ for all i. This assumption rules out interference between units and excludes the presence of multiple versions of the treatment that may lead to different potential outcomes.*

Assumption 2 (Strong Ignorability)**.** *Let $X_i \in \mathbb{R}^p$ denote a vector of pre-treatment covariates. Then:*

$$(Y_i(1), Y_i(0)) \perp\!\!\!\perp D_i \mid X_i, \tag{3}$$

$$0 < \mathbb{P}(D_i = 1 \mid X_i = x) < 1 \quad \textit{for all } x. \tag{4}$$

The first condition, often referred to as unconfoundedness, requires that all variables jointly influencing treatment assignment and potential outcomes are observed. The second condition, known as overlap or positivity, ensures that each unit has a non-zero probability of receiving each treatment level.

Under assumptions 1 and 2, the ATE is identified from the observed data distribution:

$$\tau = \mathbb{E}_X \left[\mathbb{E}[Y_i \mid D_i = 1, X_i] - \mathbb{E}[Y_i \mid D_i = 0, X_i] \right]. \tag{5}$$

This expression highlights the central role of conditioning on appropriate covariates in order to remove confounding bias.

While the potential outcomes framework defines the estimand, Structural Causal Models (SCMs) provide the graphical criteria for selecting the covariate set X_i providing a complementary perspective based on graphical representations. An SCM is defined by a set of endogenous variables $\boldsymbol{V}$, exogenous background variables $\boldsymbol{U}$, and a set of structural functions $\boldsymbol{f}$ such that $v_j = f_j(\mathrm{pa}(v_j), u_j)$, where $\mathrm{pa}(v_j)$ denotes the parents of v_j in the corresponding Directed Acyclic Graph (DAG) $\mathcal{G}$. The graphical structure encodes assumptions about causal relationships, while the structural equations specify how variables are generated.

Definition 1 (Backdoor Criterion)**.** *A set of variables X satisfies the backdoor criterion relative to (D, Y) if:*

1. *No element of X is a descendant of D;*
2. *X blocks every path between D and Y that contains an arrow pointing into D.*

Theorem 1 (Pearl, 1995)**.** *If a set of variables X satisfies the backdoor criterion relative to (D, Y), then the causal effect of D on Y is identifiable and given by the adjustment formula:*

$$\mathbb{P}(Y \mid \mathrm{do}(D = d)) = \int \mathbb{P}(Y \mid D = d, X = x)\, d\mathbb{P}(X = x). \tag{6}$$

3.1 Role of LLMs within the Causal Framework

Within this formal structure, LLMs can be interpreted as operators that map unstructured data into variables that enter the causal model. Specifically, given text data W_i, an LLM produces representations or annotations that serve as proxies for treatment variables, outcomes, or covariates. This mapping can be written abstractly as

$$(W_i) \longmapsto (D_i, Y_i, X_i),$$

where the right-hand side denotes variables used in the causal analysis. In other words, LLMs interface with these formal structures in three specific ways: (1) mapping raw text W_i to the structural variables D_i, Y_i, X_i (Measurement); (2) estimating the joint distribution $\mathbb{P}(\mathcal{G})$ over the space of

DAGs (Discovery); and (3) estimating the nuisance functions $\mu_d(X) = \mathbb{E}[Y_i \mid D_i = d, X_i = x]$ and $\pi(x) = \mathbb{P}(D_i = 1 \mid X_i = x)$ (Estimation).

A key challenge is that this mapping is inherently imperfect. Unlike classical measurement error models, where errors are often assumed to be independent of the true variable, LLM-generated errors may depend on both observed and unobserved features of the data. This dependence can introduce biases that interact with the identification assumptions stated above. For example, if measurement error depends on treatment status or latent confounders, the ignorability assumption may be violated even when it holds for the true variables.

Therefore, integrating LLMs into causal inference requires not only statistical estimation techniques but also a careful analysis of how measurement processes affect identification. The subsequent sections of the paper develop formal results that characterize these interactions and provide conditions under which valid inference remains possible.

4 LLMs as Measurement Instruments: Design-Based Supervised Learning

Consider a setting in which the true outcome $Y_i \in \mathbb{R}$ is a latent construct embedded in text W_i. Direct manual annotation of Y_i is typically infeasible at scale due to cost and time constraints. Instead, an LLM is used to construct a surrogate variable $\tilde{Y}_i = f_{\text{LLM}}(W_i)$, which serves as a proxy for the unobserved outcome.

A naive approach replaces Y_i with $\tilde{Y}_i$ in the estimation of the ATE. However, this substitution implicitly assumes that the measurement error $\varepsilon_i = \tilde{Y}_i - Y_i$ is either negligible or mean-independent of treatment and covariates. In realistic applications, this assumption is unlikely to hold. LLM-based annotations may exhibit systematic biases that depend on linguistic structure, semantic ambiguity, or contextual features of the text. In particular, if $\mathbb{E}[\varepsilon_i \mid D_i] \neq 0$, then the naive estimator

$$\hat{\tau}_{\text{naive}} = \mathbb{E}[\tilde{Y}_i \mid D_i = 1] - \mathbb{E}[\tilde{Y}_i \mid D_i = 0]$$

is asymptotically biased.

4.1 4.1 Prompt Sensitivity and Aleatory Measurement Error

While the mapping $\tilde{Y}_i = f_{LLM}(W_i)$ is often treated as a deterministic operator, practical implementation reveals that f_{LLM} is highly sensitive to the specific prompt template $\mathcal{P}$ [Ma et al., 2024, Liu et al., 2025]. We formalize this by redefining the surrogate as a function of both the text and the prompt:

$$\tilde{Y}_{i,\mathcal{P}} = f_{LLM}(W_i; \mathcal{P}) \tag{7}$$

Minor perturbations in $\mathcal{P}$ (e.g., lexical substitutions or changes in few-shot examples) can induce a prompt-dependent error component $\eta(\mathcal{P})$, such that the total measurement error becomes $\epsilon_i = \tilde{Y}_{i,\mathcal{P}} - Y_i$ [Ma et al., 2024].

If $\mathbb{E}[\tilde{Y}_{i,\mathcal{P}}] \neq \mathbb{E}[\tilde{Y}_{i,\mathcal{P}'}]$ for two semantically equivalent prompts $\mathcal{P}$ and $\mathcal{P}'$, the resulting estimator suffers from prompt instability [Liu et al., 2025]. This stochasticity suggests that f_{LLM} should be modeled as a conditional distribution $\mathbb{P}(\tilde{Y}_i|W_i, \mathcal{P})$ rather than a stable function. Failure to account for this variance may lead to spurious causal discoveries if the prompt sensitivity correlates with the treatment D_i or latent features of the text W_i.

4.2 Design-based Supervised Learning (DSL)

To address this issue, Egami et al. [2024] formalized Design-based Supervised Learning (DSL) which combines a small labeled sample with large-scale LLM predictions. Let $S_i \in \{0, 1\}$ denote a sampling indicator such that $S_i = 1$ if unit i belongs to a manually annotated subset of size n, and $S_i = 0$ otherwise. Let $\rho = \mathbb{P}(S_i = 1)$ denote the sampling probability.

Define an imputation model for the true outcome $g(X_i, D_i, \tilde{Y}_i; \theta)$, parameterized by $\boldsymbol{\theta}$, which is trained on the labeled sample S. The DSL estimator for the mean potential outcome under treatment $d \in \{0, 1\}$ is given by

$$\hat{\mu}_d^{\text{DSL}} = \frac{1}{N} \sum_{i=1}^{N} \left[g(X_i, d, \tilde{Y}_i; \hat{\theta}) + \frac{S_i}{\rho} \frac{\mathbb{I}(D_i = d)}{\hat{\pi}(d \mid X_i)} \left(Y_i - g(X_i, d, \tilde{Y}_i; \hat{\theta}) \right) \right]$$

where $\hat{\pi}(d \mid X_i)$ is the estimated propensity score.

This estimator combines outcome regression with inverse probability weighting. Its defining property is double robustness: consistency is achieved if either the imputation model $g(\cdot)$ or the propensity score $\pi(d \mid X_i)$ is correctly specified.

The presence of $\tilde{Y}_i$ in the imputation function allows the estimator to exploit the predictive structure captured by the LLM while correcting its systematic biases using the labeled subset. In this sense, DSL can be interpreted as a calibration mechanism that anchors large-scale predictions to a smaller set of ground-truth observations.

A further insight arises from examining the decomposition

$$\mathbb{E}[\tilde{Y}_i \mid D_i] = \mathbb{E}[Y_i \mid D_i] + \mathbb{E}[\varepsilon_i \mid D_i].$$

The second term captures the bias introduced by the measurement process. If ε_i depends on latent features correlated with D_i, then even adjustment on observed covariates may fail to eliminate bias. The DSL correction term compensates for this bias through the residual component involving $Y_i - g(\cdot)$.

Finally, the validity of the estimator relies on the sampling mechanism for S_i. If the labeled subset

is not representative of the full population, additional bias may be introduced. Therefore, random sampling or carefully designed stratification schemes are essential for ensuring the consistency of DSL estimators. This is formally established the following result.

Theorem 2 (Double Robustness of DSL)**.** *Under Assumptions 1 and 2, and assuming $\rho > 0$, the DSL estimator $\hat{\mu}_d^{DSL}$ is consistent for $\mathbb{E}[Y_i(d)]$ if either:*

1. *The imputation model is correctly specified: $plim_{n\to\infty} g(X_i, d, \tilde{Y}_i; \hat{\boldsymbol{\theta}}) = \mathbb{E}[Y_i \mid D_i = d, X_i, \tilde{Y}_i]$, OR*
2. *The sampling probability ρ and propensity score $\pi(d \mid X_i)$ are known or correctly specified.*

Proof Sketch. By the law of iterated expectations, if condition (2) holds, the expectation of the second term in the summand corrects exactly for the asymptotic bias of the first term. Specifically, $\mathbb{E}\left[\frac{S_i}{\rho}\right] = 1$. The term inside the parenthesis is the residual $Y_i - \hat{Y}_i$. The inverse probability weighting $\frac{\mathbb{I}(D_i=d)}{\pi(d|X_i)}$ ensures the expectation is taken over the marginal distribution of X_i. If condition (1) holds, the residual $Y_i - g(\cdot)$ has expectation zero conditional on D_i and X_i, rendering the second term zero in expectation, while the first term consistently estimates the mean. □

This formalization proves that by anchoring the LLM's high-throughput predictions $\tilde{Y}_i$ with a randomly sampled ground-truth subset, researchers can eliminate the structural bias introduced by LLM hallucinations or demographic skew, achieving root-n consistency for the causal parameter.

5 LLM-Assisted Causal Discovery as Bayesian Structure Learning

Causal discovery seeks to recover the underlying directed acyclic graph (DAG) G^* from observational data, i.e., an observational joint distribution $\mathbb{P}(\boldsymbol{V})$. In general, observational data identify only the Markov equivalence class (MEC) of G^*, rather than a unique graph.

Definition 2 (Markov Equivalence Class)**.** *Two DAGs $\mathcal{G}_1$ and $\mathcal{G}_2$ are Markov equivalent if they entail the exact same set of conditional independence (CI) relations among the variables $\boldsymbol{V}$. The set of all DAGs Markov equivalent to $\mathcal{G}^*$ is its Markov Equivalence Class (MEC).*

Theorem 3 (Verma and Pearl, 1990)**.** *Two DAGs are Markov equivalent if and only if they have the same skeleton (underlying undirected graph) and the same set of v-structures (colliders $A \to B \leftarrow C$ where A and C are not adjacent).*

Constraint-based algorithms (e.g., PC, FCI) rely on a CI oracle to identify the MEC. Because observational data alone cannot distinguish between DAGs within the same MEC, the true graph $\mathcal{G}^*$ remains unidentifiable without experimental interventions or parametric assumptions (e.g., LiNGAM).

LLMs resolve this fundamental unidentifiability by acting as an informative prior over the space of DAGs. We formalize this as a Bayesian structure learning problem. Let $\mathcal{D}$ be the observational data.

The posterior probability of a graph $\mathcal{G}$ is given by Bayes' rule:

$$\mathbb{P}(\mathcal{G} \mid \mathcal{D}) \propto \mathbb{P}(\mathcal{D} \mid \mathcal{G})\mathbb{P}_{LLM}(\mathcal{G}) \tag{8}$$

The likelihood term is computed using standard scoring functions such as BIC or Bayesian Dirichlet scores. The critical innovation is the prior $\mathbb{P}_{\text{LLM}}(G)$ that encodes semantic relationships extracted from the LLM.

For each ordered pair $(A, B) \in \boldsymbol{V}$ of variables adjacent in the MEC skeleton, the LLM is queried to output a probability $p_{A\to B} = \mathbb{P}_{LLM}(A \to B)$ based on its semantic knowledge base and representing the plausibility of a directed edge. Assuming independence across edges, the prior over the entire graph $\mathcal{G}$ factorizes as

$$\mathbb{P}_{\text{LLM}}(G) = \prod_{(i,j)\in E(G)} p_{i\to j} \prod_{(i,j)\notin E(G)} (1 - p_{i\to j}).$$

By integrating $\mathbb{P}_{LLM}(\mathcal{G})$ into greedy search algorithms (e.g., Greedy Equivalence Search, GES), this formulation effectively breaks symmetries within the MEC by favoring edges that align with semantic knowledge. However, the independence assumption underlying this factorization is strong and may not hold in realistic settings.

A critical challenge in treating $\mathbb{P}_{LLM}(\mathcal{G})$ as an independent prior is the risk of data contamination. Because LLMs are pre-trained on massive corpora that likely include the very social science literature the researcher seeks to replicate, the "prior" knowledge may be a reflection of the observational data itself rather than independent domain theory [Ma et al., 2024].

We formalize this circularity risk as a dependency between the LLM prior and the observational data $\mathcal{D}$. If $\mathcal{D}$ (or findings derived from it) is present in the training set $\mathcal{T}$ of the model, then $\mathbb{P}_{LLM}(\mathcal{G}|\mathcal{T})$ is not a true prior but a posterior-weighted distribution. This violates the Bayesian assumption of independence between the prior and the likelihood:

$$\mathbb{P}(\mathcal{G}|\mathcal{D}) \not\propto \mathbb{P}(\mathcal{D}|\mathcal{G})\mathbb{P}_{LLM}(\mathcal{G}) \tag{9}$$

In such cases, the LLM may "hallucinate" high confidence in an edge $A \to B$ simply because it has seen the correlation reported in previous studies, leading to over-confident and potentially biased structure recovery [Liu et al., 2025].

An important limitation of the LLM-assisted causal discovery as Bayesian structure learning is that LLM-derived priors reflect statistical regularities in language rather than causal mechanisms. As a result, they may encode spurious correlations or biases present in the training corpus. Consequently, the incorporation of such priors should be viewed as introducing subjective information into the discovery process.

A useful extension is to introduce a tuning parameter λ controlling the strength of the prior:

$$\mathbb{P}(G \mid D) \propto \mathbb{P}(D \mid G)\, \mathbb{P}_{\text{LLM}}(G)^{\lambda}.$$

Varying λ allows practitioners to assess the sensitivity of the inferred graph to the LLM-based prior.

6 Formal Requirements for LLM-Proposed Instrumental Variables

When Assumption 2 (Unconfoundedness) fails due to unobserved confounders $\boldsymbol{U}_i$, the ATE is no longer identified via covariate adjustment. Instrumental Variables (IV) provide an alternative identification strategy.

Definition 3 (Valid Instrumental Variable)**.** *A variable Z_i is a valid instrument for the causal effect of D_i on Y_i if it satisfies three conditions:*

1. *Relevance: $Cov(Z_i, D_i) \neq 0$.*
2. *Independence (Exogeneity): $Z_i \perp\!\!\!\perp \boldsymbol{U}_i$.*
3. *Exclusion Restriction: $Y_i(d, z) = Y_i(d)$ for all d, z. The instrument affects the outcome only through the treatment.*

Under these conditions, and assuming linear structural equations, the Local Average Treatment Effect (LATE) is identified by the Wald estimator:

$$\tau_{LATE} = \frac{\text{Cov}(Y_i, Z_i)}{\text{Cov}(D_i, Z_i)} \tag{10}$$

Discovering variables that satisfy Conditions 2 and 3 is a formidable challenge, as these are untestable assumptions that rely entirely on domain theory. Sheth et al. [2026] formalize the use of multi-agent LLM systems to propose candidate IVs.

Let $\mathcal{K}_{LLM}$ denote the latent knowledge graph encoded within the LLM's weights. The "Generator" agent executes a search over $\mathcal{K}_{LLM}$ to find a node Z such that there exists a directed path $Z \to \cdots \to D$. The "Critic" agent then evaluates Z against Conditions 2 and 3 by searching for any paths $Z \to \cdots \to U \to Y$ (violating exogeneity) or $Z \to \cdots \to Y$ not passing through D (violating exclusion).

This adversarial process can be formalized as a constraint satisfaction problem over the semantic space. Only candidate variables $\hat{Z}$ that survive the Critic's structural counter-arguments are retained. The final validation of Condition 1 (Relevance) is executed empirically by estimating the first-stage regression $D_i = \alpha_0 + \alpha_1 \hat{Z}_i + \boldsymbol{\beta}^T X_i + \nu_i$ and testing the null hypothesis $H_0 : \alpha_1 = 0$ using the observational data.

7 Doubly Robust Estimation under Text Confounding

Consider the scenario where the treatment D_i is a latent construct (e.g., "authoritarian tone") embedded within a text document W_i, and the outcome is Y_i. The text W_i also contains latent confounders $\boldsymbol{U}_i$ (e.g., "topic" or "length") that influence both the tone and the outcome. This is the problem of inference-time text confounding [Ma et al., 2025].

We utilize LLMs to map $W_i \mapsto (\hat{D}_i, \hat{\boldsymbol{U}}_i)$. Because $\hat{\boldsymbol{U}}_i$ is high-dimensional, non-parametric adjustment via matching or stratification suffers from the curse of dimensionality. We therefore turn to Double/Debiased Machine Learning (DML) [Chernozhukov et al., 2018].

Let the structural model be partially linear:

$$Y_i = \theta_0 D_i + g_0(\boldsymbol{U}_i) + \zeta_i, \quad \mathbb{E}[\zeta_i \mid D_i, \boldsymbol{U}_i] = 0 \tag{11}$$

$$D_i = m_0(\boldsymbol{U}_i) + V_i, \quad \mathbb{E}[V_i \mid \boldsymbol{U}_i] = 0 \tag{12}$$

where θ_0 is the causal parameter of interest, and g_0, m_0 are high-dimensional nuisance functions.

The DML procedure splits the sample into K folds. For each fold k, we use the complement $\mathcal{I}_k^c$ to train machine learning models (e.g., LLM-based regressors) to estimate the nuisance functions:

$$\hat{g}_0(\boldsymbol{u}) = \hat{\mathbb{E}}[Y_i \mid \boldsymbol{U}_i = \boldsymbol{u}] \tag{13}$$

$$\hat{m}_0(\boldsymbol{u}) = \hat{\mathbb{E}}[D_i \mid \boldsymbol{U}_i = \boldsymbol{u}] \tag{14}$$

We then compute the orthogonalized residuals for the units in fold k:

$$\tilde{Y}_i = Y_i - \hat{g}_0(\boldsymbol{U}_i) \tag{15}$$

$$\tilde{D}_i = D_i - \hat{m}_0(\boldsymbol{U}_i) \tag{16}$$

The debiased estimator $\hat{\theta}$ is obtained by regressing $\tilde{Y}_i$ on $\tilde{D}_i$:

$$\hat{\theta} = \left(\frac{1}{N} \sum_{i=1}^{N} \tilde{D}_i^2 \right)^{-1} \frac{1}{N} \sum_{i=1}^{N} \tilde{D}_i \tilde{Y}_i \tag{17}$$

Theorem 4 (Neyman Orthogonality)**.** *The score function $\psi(W; \theta, g, m) = (Y - \theta D - g(U))(D - m(U))$ satisfies the Neyman orthogonality condition:*

$$\frac{\partial}{\partial \eta} \mathbb{E}\left[\psi(W; \theta_0, \eta_0 + r(\eta - \eta_0))\right]\Big|_{r=0} = 0 \tag{18}$$

where $\eta = (g, m)$ represents the nuisance parameters.

Proof Sketch. The Gateaux derivative of the expected score with respect to the nuisance parameter

g in the direction $\tilde{g}$ is $\mathbb{E}[-\tilde{g}(U)(D - m_0(U))]$. Since $m_0(U) = \mathbb{E}[D \mid U]$, by the law of iterated expectations, $\mathbb{E}[\tilde{g}(U)(D - \mathbb{E}[D \mid U])] = \mathbb{E}[\tilde{g}(U)\mathbb{E}[D - \mathbb{E}[D \mid U] \mid U]] = 0$. A symmetric argument holds for perturbations in m. □

Theorem 4 guarantees that first-order estimation errors in the LLM-derived nuisance functions $\hat{g}_0$ and $\hat{m}_0$ do not propagate to the estimator $\hat{\theta}$. As long as the product of the convergence rates of the nuisance estimators is $o_p(N^{-1/2})$, $\hat{\theta}$ achieves $\sqrt{N}$-consistency and asymptotic normality, permitting valid construction of confidence intervals for the text-based treatment effect.

8 Counterfactual Auditing of LLM Fairness

The final component of this framework applies causal inference to audit the LLMs themselves. Traditional fairness metrics, such as Demographic Parity ($\mathbb{P}(\hat{Y} = 1 \mid A = 1) = \mathbb{P}(\hat{Y} = 1 \mid A = 0)$ where A is a sensitive attribute), are purely associational. They fail to distinguish between true algorithmic bias and background correlations present in the training distribution.

We formalize LLM fairness using the causal concept of Counterfactual Fairness [Amiri-Margavi et al., 2026].

Definition 4 (Counterfactual Fairness)**.** *Let A be a sensitive attribute (e.g., race, gender), $\boldsymbol{X}$ be a vector of non-sensitive attributes, and $\hat{Y}_{LLM}$ be the model's output. The LLM is counterfactually fair if, for any context $X = x$ and any individual i:*

$$\mathbb{P}(\hat{Y}_{LLM}(A \leftarrow a) = y \mid \boldsymbol{X} = x, A = a) = \mathbb{P}(\hat{Y}_{LLM}(A \leftarrow a') = y \mid X = x, A = a) \tag{19}$$

for all possible values a, a' of the sensitive attribute.

This definition explicitly utilizes the *do*-calculus intervention operator (denoted here by the assignment $\leftarrow$). It requires that the probability distribution of the LLM's output remains invariant if we were to counterfactually alter the sensitive attribute A, holding all other variables constant.

To estimate the Counterfactual Fairness Metric (CFM), we construct paired prompts. Let $P_i(a)$ be a prompt describing unit i with sensitive attribute set to a, and $P_i(a')$ be the identical prompt with the attribute set to a'. The individual-level causal bias is $\delta_i = \hat{Y}_{LLM}(P_i(a)) - \hat{Y}_{LLM}(P_i(a'))$. The Average Causal Bias (ACB) is estimated as:

$$\widehat{ACB} = \frac{1}{N} \sum_{i=1}^{N} \left(\hat{Y}_{LLM}(P_i(a)) - \hat{Y}_{LLM}(P_i(a')) \right) \tag{20}$$

Because the counterfactuals are generated perfectly by the auditor (i.e., the text of the prompt is exactly controlled), the ignorability assumption holds by design. Therefore, $\widehat{ACB}$ is an unbiased estimator of the true causal effect of the sensitive attribute on the LLM's decision-making process. If

$\widehat{ACB}$ is statistically distinguishable from zero, the model exhibits structural bias requiring mitigation before deployment in downstream causal inference tasks.

9 Conclusion

The integration of Large Language Models into empirical social science necessitates a transition from heuristic text processing to rigorous mathematical formalization. This manuscript has demonstrated that LLMs can be seamlessly embedded into the potential outcomes framework and structural causal models.

By defining the asymptotic properties of design-based supervised learning, we established the conditions under which non-random LLM measurement error can be neutralized. By formalizing causal discovery as Bayesian structure learning, we showed how LLM priors resolve Markov equivalence classes. By proving the Neyman orthogonality of score functions under text confounding, we guaranteed the root-N consistency of doubly robust estimators utilizing LLM embeddings. Finally, we defined counterfactual fairness using the *do*-calculus to isolate true algorithmic bias from observational correlation.

The synthesis of generative AI and causal inference does not relax the stringent requirements of identification; rather, it provides powerful new instruments to satisfy them. As the capabilities of LLMs expand, grounding their application in precise mathematical theorems will remain the essential safeguard for scientific validity.

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